

Lemma: R 和 R' 是环, $f: R \rightarrow R'$ 是环同态, 则对 $\forall n \in \mathbb{Z}_{\geq 0}$,

$$\forall x \in R, \text{ 都有 } f(x^n) = (f(x))^n$$

Proof: 当 $n=0$ 时, $f(x^0) = f(1_R) = 1_{R'} = (f(x))^0$

当 $n \in \mathbb{Z}_{\geq 1}$ 时, 由数学归纳法易得 $f(x^n) = (f(x))^n$

Lemma: R 和 R' 是环, $f: R \rightarrow R'$ 是环同态, 则对 $\forall n \in \mathbb{Z}$,

$$\forall x \in R^\times, \text{ 都有 } f(x^n) = (f(x))^n.$$

Proof: 当 $n \in \mathbb{Z}_{\geq 0}$ 时上面已证. 当 $n \in \mathbb{Z}_{\leq -1}$ 时, $-n \in \mathbb{Z}_{\geq 1}$

$$\therefore f(x^n) = f(x^{-(-n)}) = f((x^{-1})^{-n}) = (f(x^{-1}))^{-n}$$

$$= ((f(x))^{-1})^{-n} = (f(x))^{-(-n)} = (f(x))^n \quad \square$$

定义(算术函数) 形如 $f: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{C}$ 的映射称为算术函数

定义(乘性算术函数) $f: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{C}$ 是算术函数, 若对 $\forall a, b \in \mathbb{Z}_{\geq 1}$

$\gcd(a, b) = 1 \Rightarrow f(ab) = f(a)f(b)$, 则称 f 是乘性算术函数

定义(Dirichlet 卷积) $f: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{C}$ 和 $g: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{C}$ 是两个算术函数,

f 和 g 的 Dirichlet 卷积 $f * g: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{C}$ 也是一个算术函数, 定义为

$$f * g: \mathbb{Z}_{\geq 1} \longrightarrow \mathbb{C}$$

$$n \longmapsto (f * g)(n) = \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} f(d)g\left(\frac{n}{d}\right)$$

定义 (常值函数 1) 定义 $1: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{C}$ 为

$$1: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{C}$$
$$n \mapsto 1$$

定义 (Euler 函数) 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 定义 $\varphi(n)$ 为不超过 n 而与 n 互素的正整数个数. $\therefore$ 对 $\forall n \in \mathbb{Z}_{\geq 1}$, $\varphi(n) \in \mathbb{Z}_{\geq 1}$

$\therefore \varphi: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{C}$ 是算术函数.

令 $F: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{C}$ 为 $F = \varphi * 1$

$$\therefore \text{对 } \forall n \in \mathbb{Z}_{\geq 1}, \text{ 有: } F(n) = (\varphi * 1)(n) = \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} \varphi(d) \mid \left(\frac{n}{d}\right)$$

$$= \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} \varphi(d) \quad \therefore \text{对 } \forall n \in \mathbb{Z}_{\geq 1}, \text{ 有: } F(n) = \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} \varphi(d)$$

我们已经证明: φ 是乘性算术函数, $1: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{C}$ 也是乘性算术函数.

$\therefore F = \varphi * 1$ 也是乘性算术函数

Lemma: 对 $\forall$ 素数 p , $\forall k \in \mathbb{Z}_{\geq 1}$, 有: $F(p^k) = p^k$.

Proof: $\because p$ 是素数 $k \in \mathbb{Z}_{\geq 1} \therefore p^k \in \mathbb{Z}_{\geq 2}$.

$$F(p^k) = \sum_{\substack{d|p^k \\ d \in \mathbb{Z}_{\geq 1}}} \varphi(d) = \sum_{i=0}^k \varphi(p^i) = \varphi(1) + \sum_{i=1}^k \varphi(p^i)$$
$$= 1 + \sum_{i=1}^k p^{i-1}(p-1)$$

$$= 1 + (p-1) \sum_{i=1}^k p^{i-1} = 1 + (p-1)(p^0 + p^1 + \dots + p^{k-1}) = p^k$$

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 有: $F(n) = n$.

Proof: $\because n \in \mathbb{Z}_{\geq 1} \quad \therefore \exists r \in \mathbb{Z}_{\geq 0}$, 相异素数 $p_1, \dots, p_r; a_1, \dots, a_r \in \mathbb{Z}_{\geq 1}$

$$\text{s.t. } n = p_1^{a_1} \dots p_r^{a_r}$$

$$\therefore F(n) = F(p_1^{a_1} \dots p_r^{a_r}) = F(p_1^{a_1}) \dots F(p_r^{a_r}) = p_1^{a_1} \dots p_r^{a_r} = n. \quad \square$$

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 有: $\sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} \varphi(d) = n$.

$$\text{proof: } n = F(n) = \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} \varphi(d) \quad \square$$

Remark: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, $\because n \geq 1 \quad \therefore 1 \leq n \quad \therefore 1$ 不超过 n .

$$\therefore \gcd(1, n) = \max\{d \in \mathbb{Z}_{\geq 1} : d|1 \text{ 且 } d|n\} = \max\{1\} = 1$$

$$\therefore 1 \text{ 与 } n \text{ 互素.} \quad \therefore \varphi(n) \geq 1 \quad \therefore \varphi(n) \in \mathbb{Z}_{\geq 1}$$

$$\therefore \text{对 } \forall n \in \mathbb{Z}_{\geq 1}, \text{ 都有 } \varphi(n) \in \mathbb{Z}_{\geq 1}$$

Lemma: G 是群, $d \in \mathbb{Z}_{\geq 1}$. 定义两个集合 $S_d := \{g \in G : \text{ord}(g) = d\}$.

$$X_d = \{x \in G : x^d = 1_G\}. \text{ 则有:}$$

$$\textcircled{1} X_d = \bigsqcup_{\substack{k|d \\ k \in \mathbb{Z}_{\geq 1}}} S_k \quad \textcircled{2} S_d \subseteq X_d$$

Proof: 对 $\forall g \in \bigsqcup_{\substack{k|d \\ k \in \mathbb{Z}_{\geq 1}}} S_k$, 有: $\exists k_0 \in \mathbb{Z}_{\geq 1}$ 且 $k_0 | d$ s.t. $g \in S_{k_0}$

$$\therefore g \in S_{k_0} = \{g \in G : \text{ord}(g) = k_0\}. \quad \therefore g \in G \text{ 且 } \text{ord}(g) = k_0$$

$$\because G \text{ 是群}, g \in G, \text{ord}(g) = k_0 \in \mathbb{Z}_{\geq 1}, d \in \mathbb{Z}_{\geq 1}, \text{ord}(g) | d$$

$$\therefore g^d = 1_G \quad \therefore g \in G \text{ 且 } g^d = 1_G \quad \therefore g \in X_d \quad \therefore \bigsqcup_{\substack{k|d \\ k \in \mathbb{Z}_{\geq 1}}} S_k \subseteq X_d$$

$$\text{对 } \forall x \in X_d, \text{ 有: } x \in G \text{ 且 } x^d = 1_G \quad \therefore \text{ord}(x) \in \mathbb{Z}_{\geq 1}.$$

$$\because G \text{ 是群}, x \in G, \text{ord}(x) \in \mathbb{Z}_{\geq 1}, d \in \mathbb{Z}_{\geq 1}, x^d = 1_G$$

$$\therefore \text{ord}(x) | d. \text{ 设 } k = \text{ord}(x) \quad \therefore k \in \mathbb{Z}_{\geq 1} \text{ 且 } k | d$$

$$\therefore S_k = \{g \in G : \text{ord}(g) = k\} \quad \therefore x \in S_k$$

$$\therefore k \in \mathbb{Z}_{\geq 1} \text{ 且 } k | d \text{ 且 } x \in S_k$$

$$\therefore x \in \bigsqcup_{\substack{k|d \\ k \in \mathbb{Z}_{\geq 1}}} S_k \quad \therefore X_d \subseteq \bigsqcup_{\substack{k|d \\ k \in \mathbb{Z}_{\geq 1}}} S_k$$

$$\therefore X_d = \bigsqcup_{\substack{k|d \\ k \in \mathbb{Z}_{\geq 1}}} S_k.$$

$$\text{且 } k_1 \neq k_2$$

对 $\forall k_1, k_2 \in \mathbb{Z}_{\geq 1}$ 满足 $k_1 | d$ 且 $k_2 | d$. 有:

$$S_{k_1} = \{g \in G : \text{ord}(g) = k_1\}, S_{k_2} = \{g \in G : \text{ord}(g) = k_2\}.$$

~~假设~~ 假设 $S_{k_1} \cap S_{k_2} \neq \emptyset$. 则 $\exists g \in S_{k_1} \cap S_{k_2}$. $\therefore g \in G$ 且 $\text{ord}(g) = k_1$ 且 $\text{ord}(g) = k_2$ 且 $k_1 \neq k_2$.

$$\because \text{ord}(g) \in \mathbb{Z}_{\geq 1} \quad \because \text{ord}(g) = \min \{k \in \mathbb{Z}_{\geq 1} : g^k = 1_G\} \quad \text{矛盾.}$$

$$\because S_{k_1} \cap S_{k_2} = \emptyset. \quad \because X_d = \bigsqcup_{\substack{k|d \\ k \in \mathbb{Z}_{\geq 1}}} S_k$$

$$\because \{x \in G : x^d = 1_G\} = \bigsqcup_{\substack{k|d \\ k \in \mathbb{Z}_{\geq 1}}} \{g \in G : \text{ord}(g) = k\}. \quad \square$$

推论: G 是群, $d \in \mathbb{Z}_{\geq 1}$, 则有:

$$\{g \in G : \text{ord}(g) = d\} \subseteq \{x \in G : x^d = 1_G\}$$

Proof: 由上-引理显然. $\square$

Lemma: G 是有限群, $|G| = n \in \mathbb{Z}_{\geq 1}$, 对 $\forall x \in G$, 有: 下列命题

等价:

① x 是 G 的生成元

② $G = \langle x \rangle$

③ $\text{ord}(x) = n$

Proof: ① $\Leftrightarrow$ ② 显然.

$$\text{②} \Rightarrow \text{①}: \because G \text{ 是群}, x \in G \quad \because \text{ord}(x) = |\langle x \rangle| = |G| = n$$

$$\text{③} \Rightarrow \text{②}: \because G \text{ 是群}, x \in G \quad \because \langle x \rangle \leq G$$

$$\because \text{ord}(x) = |\langle x \rangle|, \text{ord}(x) = n \quad \because |\langle x \rangle| = n.$$

$$\because \langle x \rangle \leq G, |G| = n, |\langle x \rangle| = n \quad \because G = \langle x \rangle \quad \square$$

Lemma: G 是群, $H \leq G$, $x \in H$. 则在群 $(G, \cdot)$ 中观察元素 x 的阶 = 在群 $(H, \cdot)$ 中观察元素 x 的阶.

Proof: $\because G$ 是群, $x \in G \quad \therefore \langle x \rangle = \{x^k \mid k \in \mathbb{Z}\}$.

$\because (H, \cdot)$ 是群, $x \in H \quad \therefore \langle x \rangle = \{x^k \mid k \in \mathbb{Z}\}$

$\therefore \text{ord}(x) = |\langle x \rangle|$ 唯一 $\square$

Lemma: G 是有限群, $|G| = n \in \mathbb{Z}_{\geq 1}$, 则有:

$$\textcircled{1}: |G| = \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} |\{g \in G : \text{ord}(g) = d\}|$$

$$\textcircled{2}: G = \bigsqcup_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} \{g \in G : \text{ord}(g) = d\}$$

Proof: 先证 $\textcircled{2}$. 右 $\subseteq$ 左显然. 对 $\forall g \in G$,

$\because G$ 是有限群, $g \in G \quad \therefore \text{ord}(g) \mid |G|$. ~~设 $d = \text{ord}(g)$~~

设 $d = \text{ord}(g) \quad \therefore d \mid |G| \quad \therefore d \mid n$ 且 $d \in \mathbb{Z}_{\geq 1}$

$\therefore g \in G, \text{ord}(g) = d, d \mid n$ 且 $d \in \mathbb{Z}_{\geq 1} \quad \therefore g \in$ 右边

$\therefore$ 左 $\subseteq$ 右 $\therefore$ 左 = 右. 右边各集合显然无交. $\textcircled{2}$ 得证.

$\textcircled{1}$ 显然可得. $\square$

$(R, +, \cdot, 0_R, 1_R)$ 是环, 则已经证明了 $(R^\times, \cdot)$ 是群.

Lemma (交换环中的 n 次单位根) $(R, +, \cdot, 0_R, 1_R)$ 是交换环,

$n \in \mathbb{Z}_{\geq 1}$, 则有: $\mu_n(R) = \{r \in R^\times : r^n = 1_R\}$ 是群 $(R^\times, \cdot)$ 的子群.

集合 $\mu_n(R)$ 中的元素称为交换环 R 中的 n 次单位根.

Proof: $\because (R, +, \cdot, 0_R, 1_R)$ 是交换环 $\therefore (R^\times, \cdot)$ 是群.

且 1_R 是群 $(R^\times, \cdot)$ 的乘法幺元. $\because n \in \mathbb{Z}_{\geq 1}$ 且 $\mu_n(R) = \{r \in R^\times : r^n = 1_R\}$

$\therefore \mu_n(R) \subseteq R^\times$. $\because 1_R \in R^\times$ 且 $1_R^n = \underbrace{1_R \cdots 1_R}_{n \text{ 个 } 1_R \text{ 相乘}} = 1_R \therefore 1_R \in \mu_n(R)$

$\therefore \mu_n(R)$ 是 $R^\times$ 的非空子集, 且 $1_{R^\times} = 1_R \in \mu_n(R)$

对 $\forall x, y \in \mu_n(R)$, 有:

$\because x \in \mu_n(R) \therefore x \in R^\times$ 且 $x^n = 1_R$

$\because y \in \mu_n(R) \therefore y \in R^\times$ 且 $y^n = 1_R$

$\therefore xy \in R^\times$ 且 $(xy)^n = \underbrace{(xy) \cdots (xy)}_{n \text{ 个 } (xy) \text{ 相乘}} = x^n \cdot y^n = 1_R \cdot 1_R = 1_R$
 $\downarrow$
用到 R 是交换环

$\therefore xy \in \mu_n(R) \therefore \mu_n(R)$ 对乘法封闭.

对 $\forall x \in \mu_n(R)$, 有: $\because x \in \mu_n(R) \therefore x \in R^\times$ 且 $x^n = 1_R$

$\therefore x^{-1} \in R^\times$ 且 $(x^{-1})^n = (x^n)^{-1} = 1_R^{-1} = 1_R \therefore x^{-1} \in \mu_n(R)$

$\therefore \mu_n(R)$ 对取逆封闭. $\therefore \mu_n(R) \leq R^\times \quad \square$

Lemma: R 和 R' 是交换环, $n \in \mathbb{Z}_{\geq 1}$, 则有: 交换环之间的同态

$R \rightarrow R'$ 给出群同态 $\mu_n(R) \rightarrow \mu_n(R')$

Proof: $\because R$ 是交换环 $\therefore \mu_n(R) = \{r \in R^\times : r^n = 1_R\} \leq R^\times$

$\because R'$ 是交换环 $\therefore \mu_n(R') = \{r \in (R')^\times : r^n = 1_{R'}\} \leq (R')^\times$

设 $f: R \rightarrow R'$ 是一个环同态. $\therefore \mu_n(R) \leq R^\times \subseteq R$

$\mu_n(R') \leq (R')^\times \subseteq R'$ $\therefore$ 定义映射 $g: \mu_n(R) \rightarrow \mu_n(R')$
 $x \mapsto f(x)$

对 $\forall x \in \mu_n(R)$, 有: $x \in R^\times$ 且 $x^n = 1_R$ $\therefore x \in R$ $\therefore g(x) = f(x) \in R'$

$\because x \in R^\times \therefore f(x) \in (R')^\times \therefore g(x) \in (R')^\times$

$\therefore (g(x))^n = (f(x))^n = f(x^n) = f(1_R) = 1_{R'} \therefore g(x) \in \mu_n(R')$

$\therefore g(\mu_n(R)) \subseteq \mu_n(R')$ 显然相等的自变量的像相等

$\therefore g: \mu_n(R) \rightarrow \mu_n(R')$ 是一个映射.

对 $\forall x, y \in \mu_n(R)$, 有: $x, y \in R^\times$ 且 $x, y \in R$

$\therefore g(xy) = f(xy) = f(x) \cdot f(y) = g(x) \cdot g(y)$

$\therefore g: \mu_n(R) \rightarrow \mu_n(R')$ 是一个群同态. $\square$

Lemma: F 是一个任意的域, $n \in \mathbb{Z}_{\geq 1}$, 则有: $|\mu_n(F)| \leq n$

Proof: $\because F$ 是域 $\therefore F$ 是交换环 $\therefore \mu_n(F) = \{x \in F^\times : x^n = 1_F\} \leq F^\times$

$\because F$ 是域 $\therefore X^n - 1_F \in F[X]$

$\because X^n - 1_F$ 是 $F[X]$ 中的一个 n 次非零多项式

$\therefore X^n - 1_F$ 在 F 中至多只有 n 个相异的根.

对 $\forall x \in F$,

若 x 是多项式 $X^n - 1_F$ 的根, 则 $x^n - 1_F = 0_F \therefore x^n = 1_F$

假设 $x = 0_F$, 则 $1_F = x^n = 0_F^n = 0_F \therefore F$ 是零环.

$\because F$ 是域 $\therefore F$ 是非零环 矛盾. $\therefore x \neq 0_F$.

$\therefore x \in F \setminus \{0_F\} = F^\times \therefore x \in F^\times$ 且 $x^n = 1_F \therefore x \in \mu_n(F)$

若 $x \in \mu_n(F)$, 则 $x \in F^\times$ 且 $x^n = 1_F \therefore x \in F$ 且 $x^n - 1_F = 0_F$

$\therefore x$ 是多项式 $X^n - 1_F$ 的根.

$\therefore$ 对 $\forall x \in F$, 有: $x \in \mu_n(F) \Leftrightarrow x$ 是多项式 $X^n - 1_F$ 的根

$\therefore |\mu_n(F)| = \text{多项式 } X^n - 1_F \text{ 的根的个数} \leq n$

$\therefore |\mu_n(F)| \leq n. \quad \square$

Lemma: G 是有限群, $|G| = n \in \mathbb{Z}_{\geq 1}$. 对 $\forall d \in \mathbb{Z}_{\geq 1}$, 集合

$\{x \in G : x^d = 1_G\}$ 至多只有 d 个元素. 则有:

① 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d | n$, 集合 $\{g \in G : \text{ord}(g) = d\}$ 或者是空集, 或者有 $\varphi(d)$ 个元素

② G 是 n 阶循环群.

Proof: $\because$ 之前已经证明了: 群中元素的阶要么是正整数 要么是 $+\infty$

对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d \mid n$, 分两种情况讨论:

(i) 集合 $\{g \in G : \text{ord}(g) = d\}$ 是空集, 则集合 $\{g \in G : \text{ord}(g) = d\}$ 是空集

(ii) 集合 $\{g \in G : \text{ord}(g) = d\}$ 不是空集, 则

$\exists a \in \{g \in G : \text{ord}(g) = d\} \quad \because a \in G$ 且 $\text{ord}(a) = d$

$\because G$ 是群, $a \in G$, $\text{ord}(a) = d \in \mathbb{Z}_{\geq 1} \therefore |\langle a \rangle| = d$ 且 $\langle a \rangle \leq G$

$\therefore \langle a \rangle = \{a^k \mid k \in \mathbb{Z}\} = \{1_G, a, a^2, \dots, a^{d-1}\}$

对 $\forall k \in \mathbb{Z}$, 有: $\because a \in G \quad \therefore a^k \in G$

$\because (a^k)^d = a^{kd} = (a^d)^k = (a^{\text{ord}(a)})^k = 1_G^k = 1_G$

$\therefore a^k \in \{x \in G : x^d = 1_G\} \quad \therefore \langle a \rangle \subseteq \{x \in G : x^d = 1_G\}$

$\because |\langle a \rangle| = d$ 且 集合 $\{x \in G : x^d = 1_G\}$ 至多只有 d 个元素

$\therefore \langle a \rangle = \{x \in G : x^d = 1_G\}$

$\because G$ 是有限群, $d \in \mathbb{Z}_{\geq 1}$

$\therefore \{g \in G : \text{ord}(g) = d\} \subseteq \{x \in G : x^d = 1_G\} = \langle a \rangle$

$\therefore \{g \in G : \text{ord}(g) = d\} \subseteq \langle a \rangle$

$\because G$ 是群, $a \in G \quad \therefore \langle a \rangle \leq G \quad \therefore (\langle a \rangle, \cdot)$ 是群.

$\because G$ 是有限群, $\langle a \rangle \leq G \quad \therefore \langle a \rangle$ 是有限群.

$\therefore (\langle a \rangle, \cdot)$ 是有限群, $|\langle a \rangle| = d \in \mathbb{Z}_{\geq 1}$, 对 $\forall x \in G$, 有:
 $\text{ord}(x) = d \iff x \in G \text{ 且 } \text{ord}(x) = d \iff x \in \langle a \rangle \text{ 且 } \text{ord}(x) = d$
 $\iff \langle a \rangle = \langle x \rangle \iff x$ 是 $\langle a \rangle$ 的生成元.

$$\therefore |\{g \in G : \text{ord}(g) = d\}| = \langle a \rangle \text{ 的生成元个数} = \varphi(d)$$

① 得证.

~~② $\therefore G$ 是群 $\therefore \bigcup_{\substack{d \in \mathbb{Z}_{\geq 1} \\ d|n}} \{x \in G : x^d = |G|\} = \bigcup_{\substack{k|n \\ k \in \mathbb{Z}_{\geq 1}}} \{g \in G : \text{ord}(g) = k\}$~~

~~$\therefore$ 对 $\forall d \in \mathbb{Z}_{\geq 1}$, 有:~~

~~② $\therefore G$ 是群, $n \in \mathbb{Z}_{\geq 1}$~~

~~$$\therefore \{x \in G : x^n = |G|\} = \bigcup_{\substack{k|n \\ k \in \mathbb{Z}_{\geq 1}}} \{g \in G : \text{ord}(g) = k\}$$~~

~~$$\therefore |\{x \in G : x^n = |G|\}| = \left| \bigcup_{\substack{k|n \\ k \in \mathbb{Z}_{\geq 1}}} \{g \in G : \text{ord}(g) = k\} \right|$$~~

~~$$= \sum_{\substack{k|n \\ k \in \mathbb{Z}_{\geq 1}}} |\{g \in G : \text{ord}(g) = k\}|$$~~

~~$$\text{对 } \forall g \in G, \therefore G \text{ 是有限群} \therefore \text{ord}(g) \mid |G| \therefore \text{ord}(g) \mid n$$~~

~~$$\therefore \text{ord}(g) \in \mathbb{Z}_{\geq 1} \text{ 且 } \text{ord}(g) \mid n \therefore g \in \{g \in G : \text{ord}(g) = \text{ord}(g)\}$$~~

~~$$\therefore |\{x \in G : x^n = |G|\}| = \sum_{\substack{k|n \\ k \in \mathbb{Z}_{\geq 1}}} |\{g \in G : \text{ord}(g) = k\}| = |G| = n$$~~

②: $\because G$ 是有限群, $|G| = n \in \mathbb{Z}_{\geq 1}$

$$\therefore |G| = \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} |\{g \in G : \text{ord}(g) = d\}|$$

$$\therefore n = \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} |\{g \in G : \text{ord}(g) = d\}| \leq \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} \varphi(d) = n$$

假设 $\{g \in G : \text{ord}(g) = n\} = \emptyset$, 则有:

$$n = \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} |\{g \in G : \text{ord}(g) = d\}|$$

$$= \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1} \\ d < n}} |\{g \in G : \text{ord}(g) = d\}| + |\{g \in G : \text{ord}(g) = n\}|$$

$$= \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1} \\ d < n}} |\{g \in G : \text{ord}(g) = d\}| + |\emptyset|$$

$$= \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1} \\ d < n}} |\{g \in G : \text{ord}(g) = d\}| \leq \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1} \\ d < n}} \varphi(d) = \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} \varphi(d) - \varphi(n)$$

$$= n - \varphi(n) \quad \therefore n \leq n - \varphi(n) \quad \therefore \varphi(n) \leq 0$$

$$\because n \in \mathbb{Z}_{\geq 1} \quad \therefore \varphi(n) \in \mathbb{Z}_{\geq 1} \quad \therefore \varphi(n) \geq 1 \quad \therefore 1 \leq \varphi(n) \leq 0.$$

$$\therefore 1 \leq 0. \text{ 矛盾. } \therefore \{g \in G : \text{ord}(g) = n\} \neq \emptyset.$$

$$\therefore \exists g \in G, \text{ s.t. } \text{ord}(g) = n. \quad \therefore G = \langle g \rangle \quad \therefore G \text{ 是 } n \text{ 阶循环群. } \square$$

Lemma: F 是域, 则 ~~F^*~~ F^* 的所有有限子群都是循环群.

Proof: $\because F$ 是域 $\therefore (F, +, \cdot, 0_F, 1_F)$ 是交换除环

$\therefore (F^*, \cdot)$ 是交换群. 设 G 是 F^* 的一个任意的有限子群.

$\because G$ 是有限群. $\therefore$ 设 $|G| = n \in \mathbb{Z}_{\geq 1}$ (注意: $1_F = 1_{F^*} = 1_G$)

又 $\forall d \in \mathbb{Z}_{\geq 1}$, 有: $\{x \in G: x^d = 1_F\} \subseteq G$, $X^d - 1_F \in F[X]$

对 $\forall x \in \{x \in G: x^d = 1_F\}$, 有: $x \in G$ 且 $x^d = 1_F$

$\therefore x \in F^*$ 且 $x^d = 1_F$ $\therefore x \in F$ 且 $x^d - 1_F = 0_F$

$\therefore x$ 是 $X^d - 1_F$ 在 F 中的根.

$\therefore \{x \in G: x^d = 1_F\} \subseteq X^d - 1_F$ 在 F 中的所有根组成的集合

$\therefore |\{x \in G: x^d = 1_F\}| \leq X^d - 1_F$ 在 F 中的根的个数 $\leq d$

$\therefore G$ 是 n 阶循环群. $\square$

Lemma: F 是有限域, 则有: F^* 是循环群.

Proof: $\because F$ 是有限域, $F^* \subseteq F$ $\therefore (F^*, \cdot)$ 是有限群.

$\therefore F^*$ 是 F^* 的有限子群 $\therefore F^*$ 是循环群

$\therefore (F^*, \cdot)$ 是循环群. $\square$

Lemma: F 是域, $n \in \mathbb{Z}_{\geq 1}$, 则有: $\mu_n(F)$ 是循环群

Proof: $\because F$ 是域 $\therefore F$ 是交换除环 $\therefore \mu_n(F)$ 是群 $(F^*, \cdot)$ 的子群

$\because F$ 是域, $n \in \mathbb{Z}_{\geq 1}$ $\therefore |\mu_n(F)| \leq n$

$\therefore \mu_n(F)$ 是群 $(F^\times, \cdot)$ 的有限子群 $\therefore \mu_n(F)$ 是循环群 $\square$

Lemma: F 是域, $n \in \mathbb{Z}_{\geq 1}$, 已经证明了 $\mu_n(F)$ 是循环群,

$\mu_n(F) = \{r \in F^\times : r^n = 1_F\}$, $\mu_n(F)$ 是群 $(F^\times, \cdot)$ 的子群:

若 $r \in \mu_n(F)$ 满足 $\text{ord}(r) = n$, 则 $\mu_n(F) = \langle r \rangle \simeq (\mathbb{Z}/n\mathbb{Z}, +)$

Proof: $\because F$ 是域, $n \in \mathbb{Z}_{\geq 1}$ $\therefore |\mu_n(F)| \leq n$.

$\because r \in \mu_n(F)$ $\therefore \langle r \rangle \leq \mu_n(F)$ $\therefore |\langle r \rangle| \leq |\mu_n(F)| \leq n$.

$\because \text{ord}(r) = n$ $\therefore n = \text{ord}(r) = |\langle r \rangle| \leq |\mu_n(F)| \leq n$

$\therefore |\mu_n(F)| = |\langle r \rangle| = n$. $\therefore \mu_n(F)$ 是 n 阶循环群.

$\therefore \langle r \rangle$ 是 n 阶循环群 且 $\langle r \rangle \leq \mu_n(F)$

$\therefore \mu_n(F) = \langle r \rangle \simeq (\mathbb{Z}/n\mathbb{Z}, +)$ $\square$

定义 (域 F 中的 n 次单位原根) F 是域, $n \in \mathbb{Z}_{\geq 1}$, 已经证明了

$\mu_n(F)$ 是循环群, $\mu_n(F) = \{r \in F^\times : r^n = 1_F\}$, $\mu_n(F)$ 是群 $(F^\times, \cdot)$

的子群. 若 $r \in \mu_n(F)$ 满足 $\text{ord}(r) = n$, 则已经证明了

$\mu_n(F) = \langle r \rangle \simeq (\mathbb{Z}/n\mathbb{Z}, +)$. 此时我们称 r 为域 F 中的 n 次单位原根.

Remark: 以后需要给出域 F 中存在 n 次单位原根的充要条件.

Lemma: G 是群, $g \in G$, $\text{ord}(g) = n \in \mathbb{Z}_{\geq 1}$, 则对 $\forall k \in \mathbb{Z}$, 有:

$$\text{ord}(g^k) = \frac{n}{\gcd(k, n)} = \frac{\text{ord}(g)}{\gcd(k, n)}$$

Proof: 当 $k=0$ 时, $g^0 = |_G$, $\text{ord}(g) = n \in \mathbb{Z}_{\geq 1}$,

$$\gcd(k, n) = \gcd(0, n) = n.$$

$$\therefore \text{ord}(g^k) = \text{ord}(g^0) = \text{ord}(|_G) = 1 = \frac{n}{n} = \frac{n}{\gcd(k, n)} = \frac{\text{ord}(g)}{\gcd(k, n)}$$

$\therefore$ 当 $k=0$ 时, 结论得证

当 $k \neq 0$ 时, 记 $d = \gcd(k, n) \in \mathbb{Z}_{\geq 1} \therefore d|k$ 且 $d|n$.

$$\text{设 } \frac{k}{d} = k_1 \in \mathbb{Z}, \quad \frac{n}{d} = n_1 \in \mathbb{Z}_{\geq 1} \quad \therefore k = dk_1, \quad n = dn_1$$

$$\therefore k \in \mathbb{Z}, \quad n \in \mathbb{Z}_{\geq 1} \quad \therefore \gcd\left(\frac{k}{d}, \frac{n}{d}\right) = 1$$

$$\therefore \gcd\left(\frac{k}{d}, \frac{n}{d}\right) = 1 \quad \therefore \gcd(k_1, n_1) = 1 \quad \therefore \gcd(n_1, k_1) = 1$$

$$\therefore g \in G, \quad k \in \mathbb{Z} \quad \therefore g^k \in G$$

$$\begin{aligned} \therefore (g^k)^{n_1} &= g^{k \cdot n_1} = g^{dk_1 \cdot n_1} = g^{k_1 \cdot dn_1} = g^{k_1 \cdot n} = (g^n)^{k_1} \\ &= (g^{\text{ord}(g)})^{k_1} = |_G^{k_1} = |_G \end{aligned}$$

$$\therefore G \text{ 是群}, \quad g^k \in G, \quad n_1 \in \mathbb{Z}_{\geq 1}, \quad (g^k)^{n_1} = |_G \quad \therefore \text{ord}(g^k) \in \mathbb{Z}_{\geq 1}$$

$$\therefore G \text{ 是群}, \quad g^k \in G, \quad \text{ord}(g^k) \in \mathbb{Z}_{\geq 1}, \quad n_1 \in \mathbb{Z}_{\geq 1}, \quad (g^k)^{n_1} = |_G$$

$$\therefore \text{ord}(g^k) \mid n_1$$

$$\because G \text{ 是群}, g^k \in G, \text{ord}(g^k) \in \mathbb{Z}_{\geq 1} \quad \therefore (g^k)^{\text{ord}(g^k)} = |G|$$

$$\therefore g^{k \cdot \text{ord}(g^k)} = |G|$$

$$\because G \text{ 是群}, g \in G, \text{ord}(g) = n \in \mathbb{Z}_{\geq 1}, k \cdot \text{ord}(g^k) \in \mathbb{Z}$$

$$g^{k \cdot \text{ord}(g^k)} = |G| \quad \therefore \text{ord}(g) \mid k \cdot \text{ord}(g^k). \quad \therefore n \mid k \cdot \text{ord}(g^k)$$

$$\therefore d n_1 \mid d k_1 \cdot \text{ord}(g^k). \quad \therefore \frac{d k_1 \cdot \text{ord}(g^k)}{d n_1} \in \mathbb{Z}$$

$$\therefore \frac{k_1 \cdot \text{ord}(g^k)}{n_1} \in \mathbb{Z} \quad \therefore n_1 \mid k_1 \cdot \text{ord}(g^k) \quad \therefore n_1 \mid \text{ord}(g^k)$$

$$\because n_1 \in \mathbb{Z}_{\geq 1}, \text{ord}(g^k) \in \mathbb{Z}_{\geq 1}, \text{ord}(g^k) \mid n_1, n_1 \mid \text{ord}(g^k)$$

$$\therefore \text{ord}(g^k) = n_1 = \frac{n}{d} = \frac{n}{\gcd(k, n)} = \frac{\text{ord}(g)}{\gcd(k, n)} \quad \square$$

直接引用的结论 1: 对 $\forall \theta_1, \theta_2 \in \mathbb{R}$, 有: $e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$,

$$\frac{e^{i\theta_1}}{e^{i\theta_2}} = e^{i(\theta_1 - \theta_2)}$$

直接引用的结论 2: 对 $\forall x \in \mathbb{R}, \forall n \in \mathbb{Z}$, 有:

$$(e^{ix})^n = e^{i(nx)} = e^{inx}$$

直接引用的结论 3 (Euler's formula) 对 $\forall x \in \mathbb{R}$, 有:

$$e^{ix} = \cos x + i \sin x.$$

Remark: 之后给出指数函数 $f(z) = e^z$ 和三角函数 $f(z) = \cos z$ 和 $f(z) = \sin z$ 对 $\forall z \in \mathbb{C}$ 的严格定义, 再证明欧拉公式.

Lemma: 对 $\forall x \in \mathbb{R}$, 有: $e^{ix} = 1 \Leftrightarrow x \in 2\pi\mathbb{Z}$

Proof: 对 $\forall x \in \mathbb{R}$, 有:

$$e^{ix} = 1 \Leftrightarrow \cos x + i \sin x = 1 \Leftrightarrow \cos x = 1 \text{ 且 } \sin x = 0$$

$$\Leftrightarrow x = 2k\pi, k \in \mathbb{Z} \Leftrightarrow x \in 2\pi\mathbb{Z} \quad \square$$